

to develop for OS X with this one card - a strong, native UNIX shell that is not based on any modification or emulation. On the other hand, Windows has been remarkably bad with developers on many points. For example, it was only in 2015 that Node finally got full fledged Windows support and that still is quite a mess, as can be seen on numerous complaints on online forums and Github. With the arrival of Bash on Windows, this is one of the many problems to be solved.

When it comes to web hosting, Linux is the leader with massive popularity, with over 68% of public websites being hosted from Linux based servers. With its efforts increasing everyday to establish itself as a credible cloud technology provider, it is only natural that Microsoft wants to push into the popularity that Linux based hosting has. Bringing native Bash support to Windows might be the card that Microsoft is playing for this purpose, with the increasing possibility of developers wanting to program on Windows due to the appeal of Bash - who knows, they might end up developing in Microsoft's own languages along the way.

Let us make one thing clear - this isn't a virtual machine, a container, or Linux software specifically compiled for Windows. With this extension, Windows 10 gets a Linux subsystem on Windows, which does remind one of the abandoned Project Astoria for making Android Apps work on Windows. For those of you who are familiar with Wine, this is somewhat like an exact opposite of that. While Wine allowed running Windows applications directly on Linux, this Linux subsystem will allow you to run Linux applications on Windows.

Actually, this is technically not Linux at all. Linux is the underlying operating system kernel, and that is not present here. Microsoft and Canonical have worked together to bring a full Ubuntu-based Bash shell that runs on the subsystem. What this subsystem does is, it allows you to run the Bash shell and the same binaries that you run on Ubuntu Linux. You may have heard the argument that the average Linux system should be called GNU/Linux because of the volume of GNU utilities and software running on the Linux kernel. What you get here with the Bash shell are those GNU utilities and software.

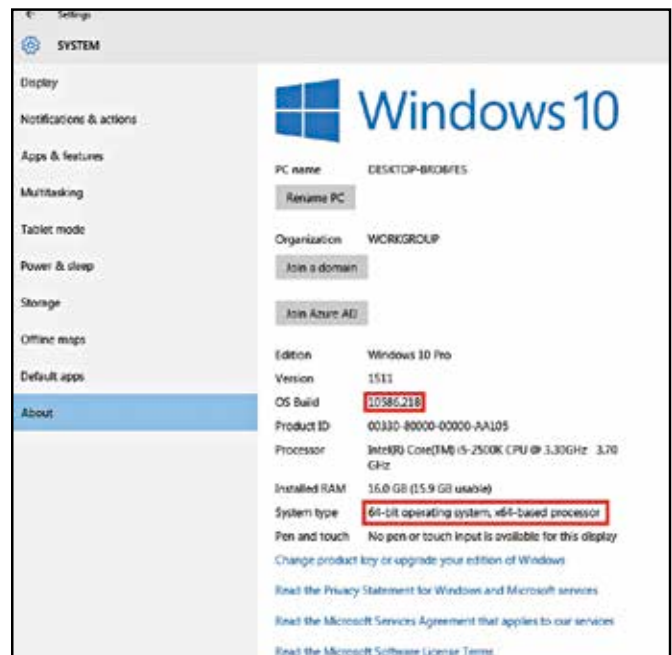
Not all of it is hunky-dory though. There are certain limitations that this implementation has. As of now, it can only deal with command line utilities. So, no server software and no graphical software. Also, even though it has access to the Windows file system, Bash commands cannot be used to automate Windows applications, and the Windows command line cannot launch Bash commands. And since it is in beta, expect some command line functions to not work.

Note- There has been a reported issue of Bash processes and applications not showing up in the Windows Task Manager or Resource monitor, even though the process can go up to using 100% of the CPU. This obviously has dire implications, as this makes Bash-enabled Windows systems the perfect targets for Bitcoin miners and Malware. This issue has been confirmed by the BashOnWindows team and is expected to be fixed by the next release.

How to get Bash on Windows

One of the first things that you need to make sure of is that the build number for Windows 10 on your system is 14316 or higher. There are two quick steps to check that :-

- On Windows 10, click or tap the **Start** button and then **Settings**.
- Navigate to **System->About** and scroll down to see the Version and Build Number fields.



In the settings app, you can see the Build number and version of your Windows installation

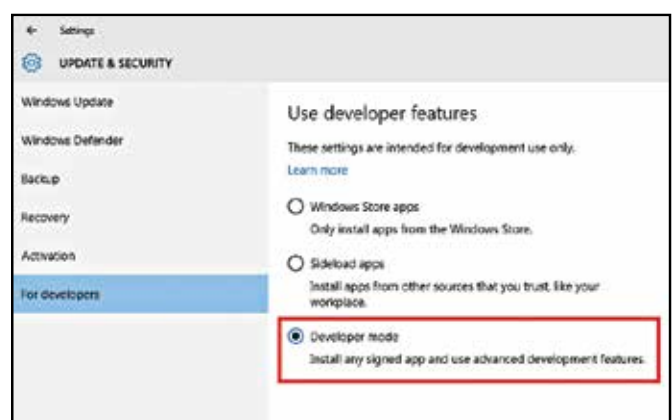
If your build number is lower than 14316, you will have to update Windows on your system before you can proceed. Once you have made sure that the build is up to date, now you need to ensure that you are part of the Insider Preview program and the same is enabled on your system. Microsoft is launching this support officially with the Anniversary build, and to try it before that, Insider Preview builds need to be enabled on your system. If it's not already enabled, here's how you can get it done - <http://dgit.in/WIPBuilds>

Also, if you are on a 32-bit version of Windows 10, now would be the time to switch to 64-bit, as this new tool works only on the 64-bit versions of Windows 10. Here's a quick how-to on getting that done - <http://dgit.in/Win3264>

Once you are sure that all the above requirements are met and you are on a correct version of Windows, follow these steps -

Step 1:

Click the **Start** button, go to the **Settings** app and navigate to **Update & Security->For Developers**. Here, tap the **"Developer Mode"** switch to turn on developer mode.



This allows you to install any signed app and advanced development features.